

Freewave

The **Freewave** is a 900 MHz spread-spectrum RF radio modem used for **short-range, high-speed** communication with a Slocum glider. When the glider is on the surface within line-of-sight of a shoreside (or boat-side) Freewave, it gives you a fast, no-cost console — fast enough that **no cable is needed** for in-lab work, so most bench communication is done over RF rather than a direct connection.

Each glider carries a **slave** modem internally; you operate a **master** modem at the dock, in the lab, or on the recovery boat. The link is the glider operator's primary backup to Iridium: short range, but free and immediate when you can reach it.

Source

Paraphrased from the *Slocum G3 Glider Operators Manual* (Rev. 1, "RF Modem Telemetry" and Appendix F, "FreeWave Configuration"), the UG2 community Slack, and the Teledyne Webb Research user forum. Appendix F itself reproduces excerpts of the FreeWave Technologies *Spread Spectrum User Manual* — for the radios themselves, refer to FreeWave's documentation. This is a condensed field reference — always defer to the official Teledyne documentation for your specific glider.

Master and slave roles

The link is **point-to-point**: one master talks to one slave at a time.

Role	Where	Factory configuration
Slave	Inside the glider, on the flight Persistor	Set to call all masters
Master	Shoreside / lab / boat	Set to talk to one specific glider

- The glider **slave is factory-set to "call all"**, so any properly configured master can reach it. Teledyne strongly recommends **never changing the slave** configuration on the glider.